

AA411

Forces Test Procedure

Team Name: ROFASO

Project Name: Damage Assessment Protocol (D.A.P.)

Sub-System Name: Plant: Forces

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1 Objective of Test Procedure

The goal of this procedure is to verify that the Seaglider forces code equations correctly calculate the lift and drag forces upon the Seaglider geometry using angle of attack (AoA, α) inputs and computational fluid dynamics (CFD) outputs.

i Scope

- Sub-system: Plant (Dynamics + Forces)
- Type: Code verification and physics model validation
- Outputs (internal): Lift, Drag, Buoyancy

2 Test Parameters and Variables

i Parameters

Parameters	Description	Value	Source
S	Surface area of glider in XY plane	0.486 m ²	CAD
m_{bat}	Battery mass	11.636 kg	ballasting data sheet
m_s	Stationary glider mass	kg	ballasting data sheet
V_{static}	Static displaced volume of glider excluding bladder	m ³	
ρ	Density	1.023 $\frac{g}{cm^3}$	log files
P	Pressure	0.03 - 85.67 decibars	log files
T	Water temperature	9°C - 11°C	log files
T_0	Ambient temperature of water surface	10°C	log files
κ	Compressibility of Seaglider	5.529e-6 decibar ⁻¹	2021 Bennett et al Paper
τ	Volumetric expansion of the Seaglider	7.05e-5 1/C°	2021 Bennett et al. Paper

ii Variables

Variable	Description	Units
Independent Variables (Control Inputs)		
t	Time stamp at which actuation states are commanded	s
VBD_cc	Volume of oil in VBD	cc
α	Angle of Attack	deg
Dependent Variables (State Variables)		
Volume	Total volume of the Seaglider	cc
q	dynamic pressure ($q = 0.5\rho v^2$)	Pa
C_L	Lift Force Coefficient	
C_D	Drag Force Coefficient	

iii Measurements

Variable	Description	Units	Measurement Method
L	Lift force	N	Forces function output
D	Drag Force	N	Forces function output
B	Buoyant Force	N	Forces function output

3 Equipment Required

Equipment/Tools	Description	Qty	Check
Laptop	At least 8 GB RAM and 256 GB SSD is required	1	
MATLAB R2024a	Simulation environment. Add ons needed: • Aerospace Toolbox • Control System Toolbox • Optimization Toolbox • Robust Control Toolbox	1	
dynamicsV3	Latest code that models the dynamics of glider	1	
ode45 solver	Solves dynamicsV3 function for discrete time steps given an current condition	1	
Parameter and Coefficients for nominal case	Input dataset	1	
Google Spreadsheet	Used to calculate and graph lift/drag force versus AoA	1	

4 Procedure: Lift and Drag Forces

1. Setup

1. Access the Google sheet file *CFD Post Processing* from the ROFASO Google Drive.
 - (a) AA - Resilient Ocean Flight Autonomy for Seaglider Operation → StarCCM+ → CFD Post Processing.
2. Find the C_L and C_D coefficients for planar flight at all AoA, e.g. $\beta = 0$.
3. Access the Google Sheet file *Forces Test - Lift and Drag*.
 - (a) AA - Resilient Ocean Flight Autonomy for Seaglider Operation → Meeting Agenda and Slides → Forces → Forces Test - Lift and Drag

Pass/Fail Criteria: Access to the test documents

2. Calculating Force of Lift and Drag from CFD

1. Open *Forces Test - Lift and Drag*
2. Input the AoA (α), C_L , and C_D (for $\beta = 0$) coefficient information into the first three columns of the Google Sheet
3. Use the equations for lift force (L) and drag force (D) from the dynamicsV3 code and calculate the L and D for each corresponding AoA

$$L = qSC_L$$

$$D = qSC_D$$

4. Graph both the lift and the drag force against AoA

3. Calculating Force of Lift and Drag from Hydrodynamic Model Equations (Bennett et al.)

1. Keep the *Forces Test - Lift and Drag* Google Sheet open
2. Using the same AoA sweep used in **Section 2**, calculate L and D using the hydrodynamic model equations

$$L = qS\alpha a$$
$$D = qS(bq^{1/4} + c\alpha^2)$$

4. Calculating Percent Difference Between CFD and HDM

1. In the same Google Sheet, calculate the percentage difference between the two forces calculated from CFD and calculated from HDM for all AoA

$$\%diff = \frac{|CFD - HDM|}{|HDM|} \cdot 100$$

Pass/Fail Criteria: The difference between the lift, drag of CFD and HDM are $\leq 3.30\%$